

Unit - 2 Statistical Techniques

Curve fitting - Suppose we have data set

$$x \rightarrow x_1, x_2, x_3, \dots, x_n$$

$$y \rightarrow y_1, y_2, y_3, \dots, y_n$$

from the above given data set we have to fit these data to the given eqn of curve

for this we have to find a best fitted curve by using Method of least sq. This is called curve fitting.

Case-I Fitting of a straight line $y = a + bx$ to the given data set.

$$y = a + bx \quad \text{--- (I)}$$

where a & b are called unknowns.

The normal eqns of (I) are:

$$\begin{aligned} \sum y &= a \sum 1 + b \sum x & \text{--- (ii)} & \times \sum \\ \sum y &= an + b \sum x \\ \sum xy &= a \sum x + b \sum x^2 & \text{--- (iii)} & \times x \end{aligned}$$

where $n \rightarrow$ Total no. of data.

On solving (ii) & (iii), we get the value of a & b . These values of a & b put in eqn of curve (I) we will get req. best fitted curve.

Ex: Fit a st. line to the following data set by using a method of least squares.

$x \rightarrow$	1	2	3	4	5
$y \rightarrow$	14	27	40	55	68

Let the st. line is: $y = a + bx$ --- (I)

Normal eqn of (I) are:

$$\sum y = an + b \sum x \quad \text{--- (ii)}$$

$$\sum xy = a \sum x + b \sum x^2 \quad \text{--- (iii)}$$

x	y	xy	x^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
<u>15</u>	<u>204</u>	<u>748</u>	<u>55</u>

$$\begin{aligned} \text{from (ii): } 204 &= 5a + 15b & \times 2 \\ 748 &= 15a + 55b \end{aligned}$$

$$\text{On solving: } 612 = 15a + 45b$$

$$\begin{aligned} 136 &= 10b \Rightarrow b = 13.6 \\ \boxed{a = 0} \\ \boxed{b = 13.6} \end{aligned}$$

$$\begin{aligned} 204 &= 5a + 15 \times 13.6 \\ &= 5a + 204 \\ 5a &= 0, a = 0 \end{aligned}$$

Q. Fit the given data set to the second degree parabola by using method of least square.

$$y = a + bx + cx^2 \quad \text{--- (1)}$$

$$\begin{aligned} \sum y &= a \sum 1 + b \sum x + c \sum x^2 \\ \sum xy &= a \sum x + b \sum x^2 + c \sum x^3 \\ \sum x^2 y &= a \sum x^2 + b \sum x^3 + c \sum x^4 \end{aligned}$$

x	y	x ²	x ³	xy	x ⁴	x ² y
1	1.8	1	1	1.8	1	1.8
2	5.1	4	8	10.2	16	20.4
3	8.9	9	27	26.7	81	80.1
4	14.1	16	64	56.4	256	225.6
5	19.8	25	125	99.0	625	495.0
15	49.7	55	225	194.1	979	822.9

$$\begin{aligned} 49.7 &= 55a + 15b + 5c \\ 194.1 &= 225a + 55b + 15c \\ 822.9 &= 979a + 225b + 55c \end{aligned} \quad \begin{cases} -0.291 \\ 1.64 \\ 0.46 \end{cases} \quad \begin{cases} a = 0.472 \\ b = 1.843 \\ c = -0.46 \end{cases}$$

$$\begin{aligned} y &= a + bx + cx^2 \\ \sum y &= an + b \sum x + c \sum x^2 \\ \sum xy &= a \sum x + b \sum x^2 + c \sum x^3 \\ \sum x^2 y &= a \sum x^2 + b \sum x^3 + c \sum x^4 \end{aligned}$$

Case-II Eqns look like polynomial.

eg- $y = ax + b$ --- (1)
Normal eqnⁿ of (1):

$$\begin{aligned} \sum xy &= a \sum x + b \sum 1 \\ \sum \frac{y}{x} &= a \sum 1 + b \sum \frac{1}{x} \end{aligned} \quad \begin{matrix} \text{--- (1)} \\ \text{--- (11)} \end{matrix}$$

(2) $y = a\sqrt{x} + \frac{b}{\sqrt{x}}$ etc.

Q.1. Use the method of least squares to fit the curve $y = \frac{C_0}{x} + C_1 \sqrt{x}$ to the following.

x	0.1	0.2	0.4	0.5	1	2
y	21	11	67	06	05	06

Normal eqn -

$$\sum \frac{y}{x} = C_0 \sum \frac{1}{x^2} + C_1 \sum \frac{1}{\sqrt{x}} \quad \text{--- (1)}$$

$$\sum y\sqrt{x} = C_0 \sum \frac{1}{\sqrt{x}} + C_1 \sum x \quad \text{--- (11)}$$

x	y	y/x	1/x	1/x ²	1/√x	√x	y√x
0.1	21	210	10	100	3.162	0.316	6.636
0.2	11	55	5	25	2.236	0.447	4.917
0.4	67	175	2.5	6.25	1.581	0.632	4.224
0.5	06	12	2	4	1.414	0.707	2.242
1	05	5	1	1	1	1	5
2	06	3	0.5	0.25	0.707	1.414	8
11	112	220	91	136.5	10.1	4.51	33.7

$$9n \text{ (i)}: 302.5 = C_0(136.5) + C_1(10.1) \text{ --- (iii)}$$

$$9n \text{ (ii)}: 33.703 = C_0(10.1) + C_1(4.2) \text{ --- (iv)}$$

$$\text{(iii)} - \text{(iv)}: \times (2.4047)$$

$$C_0(136.5) + C_1(10.1) = 302.5$$

$$C_0(24.288) + C_1(10.1) = 81.048$$

$$(112.212) C_0 = 221.452$$

$$C_0 = 1.974$$

$$C_1 = 3.278$$

$$\therefore \text{Best fit curve } y = 1.974 + 3.278\sqrt{x}$$

Q. Fit the given data set to the curve $xy = ax + b$

$x \rightarrow$	1	3	5	7	9	10
$y \rightarrow$	36	29	28	26	24	15

$$\text{Normal eqn: } \sum xy = a \sum x^2 + b \sum y \text{ --- (i)}$$

$$\sum xy = a \sum x + b \sum x \text{ --- (ii)}$$

x	y	xy	x^2	x^2y
1	36	36	1	36
3	29	87	9	261
5	28	140	25	700
7	26	182	49	1274
9	24	216	81	1944
10	15	150	100	1500
35	158	811	265	5715

9n (i):

$$5715 = (265)a + (158)b \text{ --- (iii)}$$

9n (ii):

$$811 = (35)a + 6b \text{ --- (iv)}$$

Case-III When the given curve is not a polynomial.

$$1) y = ae^{bx}, y = ab^x, \text{ or } y = k (\text{const.})$$

$$\text{Take } y = ae^{bx} \text{ --- (i)}$$

Taking $\log$ both sides on both sides.

$$\log_{10} y = \log_{10} a e^{bx}$$

$$= \log_{10} a + \log_{10} e^{bx}$$

$$= \log_{10} a + bx \log_{10} e$$

$$\text{Let } \log_{10} y = Y$$

$$\log_{10} a = A \Rightarrow$$

$$b \log_{10} e = B$$

$$x = X$$

$$\text{know, } Y = A + Bx \text{ (Eqn of st. line)}$$

$$\text{Normal eqn - } \sum Y = An + B \sum x$$

$$\sum XY = A \sum x + B \sum x^2$$

On solving, we get A, B .

$$a = 10^A$$

$$b = \frac{B}{\log_{10} e}$$

Case - When the format of the curve is not a polynomial.

Q.1. Fit the curve in form $y = ae^{bx}$ to the following data set -

$x \rightarrow$	1	5	7	9	12
$y \rightarrow$	10	15	12	15	21

$$y = ae^{bx} \quad \text{--- (I)}$$

Taking log both sides on base 10.

$$\log_{10} y = \log_{10} ae^{bx}$$

$$= \log_{10} a + \log_{10} e^{bx}$$

$$= \log_{10} a + bx \log_{10} e$$

Let $\log_{10} y = Y$, $\log_{10} a = A$, $b \log_{10} e = B$,
 $x = X$

$$Y = A + BX \quad \text{--- (II)}$$

Normal eqn (II) are:

$$\sum Y = An + B \sum x \quad \text{--- (III)}$$

$$\sum XY = A \sum x + B \sum x^2 \quad \text{--- (IV)}$$

$X=x$	Y	$Y = \log_{10} y$	x^2	XY
1	10	1	1	1
5	15	1.1761	25	5.8805
7	12	1.0792	49	7.5544
9	15	1.1761	81	10.5849
12	21	1.3222	144	15.8664
34	73	5.7536	300	11.0000

from (III) & (IV):

$$5.7536 = 5A + 34B \quad \times 34$$

$$40.8862 = 34A + 300B \quad \times 5$$

$$195.6224 = 170A + 1156B$$

$$204.431 = 170A + 1500B$$

$$8.8086 = 344B$$

$$B = 0.025606$$

$$5.7536 = 5A + 0.8704$$

$$4.8832 = 5A$$

$$A = 0.97664$$

$$A = \log_{10} a$$

$$a = 10^A \Rightarrow a = 10^{0.9766}$$

$$a = 9.4755$$

$$B = b \log_{10} e \Rightarrow b = \frac{B}{\log_{10} e} = \frac{B \log_e 10}{1}$$

$$b = 0.0256 \times \ln 10$$

$$= 0.0256 \times 2.3026$$

$$= 0.0589$$

$$y = 9.4755 e^{0.0589x}$$

Q. Fit the curve $PV^x = k$ to the data:

P (Kg/cm ²)	0.5	1	1.5	2	2.5	3
V (litres)	1620	1000	750	620	520	460

10m *

$$PV^Y = K \quad \begin{cases} y = f(x) \\ v = f(p) \end{cases}$$

$$(v^Y)^{1/Y} = \left(\frac{K}{P}\right)^{1/Y}$$

$$V = \frac{K^{1/Y}}{P^{1/Y}}$$

$$\log_{10} V = \log_{10} K^{1/Y} - \log_{10} P^{1/Y}$$

$$\log_{10} V = \frac{1}{Y} \log_{10} K - \frac{1}{Y} \log_{10} P$$

$$y = a - bx \quad \text{--- (I)}$$

$$y = \log_{10} V \Rightarrow a = \frac{1}{Y} \log_{10} K$$

$$b = \frac{1}{Y}$$

$$x = \log_{10} P$$

$$PV^{1.4225} = 18462.48$$

$$y = a - bx$$

normal eqn:

$$\sum y = a \sum 1 - b \sum x \quad \text{--- (II)}$$

$$\sum xy = a \sum x - b \sum x^2 \quad \text{--- (III)}$$

DATE

x	y
0.5	1620
1	1000
1.5	750
2	620
2.5	520
3	460

Method of least square

$$y = a + bx \quad \text{--- (I)}$$

$$L = \text{Total error} = (y_1 - a - bx_1)^2 + (y_2 - a - bx_2)^2 + \dots$$

$$E = \sum_{i=1}^n (y_i - a - bx_i)^2 \quad \text{--- (II)}$$

$$\frac{\partial E}{\partial a} = 0, \quad \frac{\partial E}{\partial b} = 0$$

$$\frac{\partial E}{\partial a} = \sum (y_i - a - bx_i) (n-1) = 0$$

$$0 = -2 \sum (y_i - a - bx_i)$$

$$\sum (y_i - a - bx_i) = 0$$

$$\sum y_i - \sum a - \sum bx_i = 0$$

$$\sum y_i = \sum a + \sum bx_i$$

$$\sum y = ax + b \sum x$$

$$\frac{\partial E}{\partial b} = \sum (y_i - a - bx_i) (-x_i)$$

$$= -2 \sum (x_i y_i - ax_i - bx_i^2)$$

$$0 = 2 \sum x_i y_i - 2 \sum ax_i - 2 \sum bx_i^2$$

$$\sum x_i y_i = a \sum x_i + b \sum x_i^2$$

Co-relation - In the bivariate distribution, if the change in one variable affects the change in the other var. the var. are said to be co-related. The co-relⁿ gives us the degree of relationship among the variables.

If the two variables deviate in the same direcⁿ i.e. if the increase (or decrease) in one var. results in a corresponding ↑ or ↓ in the var., then the co-relⁿ is said to be direct or true corrⁿ.

(+ve) +1
(-ve) -1

eg- Income & expenditure. (+ve/dir. corrⁿ).

If the two var. deviate in opp. direcⁿ i.e. if the ↑ or ↓ in one var. results in corresp. ↓ or ↑ in the other var. then the corr. is said to be inverse or -ve corr.

Co-relⁿ is said to be perfect if the dev. in one var. is followed by corr. equal proportional in the other var.

Karl Pearson's coeff. of co-relⁿ -

$$r_{xy} = r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

where $\sigma_x \rightarrow$ S.D. of x
 $\sigma_y \rightarrow$ " " y

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y}$$

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

Q.1. Find the coeff. of co-relⁿ for the foll. data by Karl Pearson method.

x	y	xy	x ²	y ²
1	8	8	1	64
3	12	36	9	144
5	15	75	25	225
7	17	119	49	289
8	18	144	64	324
10	20	200	100	400
34	90	582	248	1446

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{6 \times 582 - 34 \times 90}{\sqrt{6 \times 48 - (34)^2} \sqrt{6 \times 44 - (90)^2}}$$

$$= \frac{3492 - 3060}{\sqrt{148 - 1156} \sqrt{8676 - 8100}}$$

$$-1 \leq r \leq 1$$

$$= \frac{432}{\sqrt{332} \sqrt{576}}$$

$$= \frac{432}{18.22 \times 24} = \frac{432}{437.28}$$

$$= 0.98792$$

The Karl Pearson coeff. of correlⁿ gives the degree of linear association b/w two variables.

Find the coeff. of co-relⁿ by KP method

x	y	xy	x ²	y ²	(y - 22)
1	2	2	1	4	
2	4	8	4	16	
3	6	18	9	36	
4	8	32	16	64	
5	10	50	25	100	
15	30	110	55	220	

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{5 \times 110 - 15 \times 30}{\sqrt{5(55) - (15)^2} \sqrt{5(220) - (30)^2}}$$

$$= \frac{550 - 450}{\sqrt{275 - 225} \sqrt{1100 - 900}}$$

$$= \frac{1000}{\sqrt{50} \sqrt{200}} = \frac{1000}{5 \times \sqrt{2} \times \sqrt{5} \times \sqrt{2}} = 1$$

Q. The following data regarding the height (y) & weight (x) of 100 students as-

$$\sum x = 15000, \quad \sum x^2 = 2272500$$

$$\sum y = 6800, \quad \sum y^2 = 4630255, \quad \sum xy = 1022250$$

Find the coeff. of co-relⁿ.

$$r_{xy} = \frac{100 \times 1022250 - 15000 \times 6800}{\sqrt{100(2272500) - (15000)^2} \sqrt{100(4630255) - (6800)^2}}$$

$$= \frac{1022250 - 1020000}{\sqrt{227250000 - 225000000} \sqrt{463025500 - 46240000}}$$

$$= \frac{2250}{\sqrt{2250000} \sqrt{62500}}$$

$$= \frac{2250}{1500 \times \sqrt{62500}} = \frac{2250}{1500 \times 250}$$

$$= \frac{225}{37500} = 0.6$$

Q-3.

x	y	xy	x^2	y^2	$y=x^2$
1	1	1	1	1	
2	4	8	4	16	
3	9	27	9	81	
4	16	64	16	256	
5	25	125	25	625	
15	55	225	55	979	

$$r = \frac{5 \times 225 - 15 \times 55}{\sqrt{5 \times 55 - (15)^2} \sqrt{5 \times 979 - (55)^2}}$$

$$= \frac{1125 - 825}{\sqrt{275 - 225} \sqrt{4895 - 3025}}$$

$$= \frac{300}{\sqrt{50} \sqrt{1870}}$$

$$= \frac{300}{7.07 \times 43.24} = \frac{300}{305.70}$$

$$= 0.9813$$

Spearman's rank co-relation coeff.

Spearman's $r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$

where, $D = R_1 - R_2$

$n \rightarrow$ total no. of data.

Note - This for is app. for no tied relⁿ.

This formula gives better result for monotonic relⁿ b/w the var.

Q-1. The marks secured by students in the selection test (x) and the proficiency test (y) are given. Find S.R.C. of co-relⁿ.

R ₁	x	y	R ₂	D = R ₁ - R ₂	D ²
9	10		R ₂	0	0
5	15	30.9		2	4
8	19	42.3		6	36
3	17	45.2		2	4
7	13	46.1		-1	1
4	16	33.8		-3	9
1	(24)	34.7		-3	9
6	14	40.4		0	0
2	(22)	35.6		-3	9
		39.5			
					72

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6(72)}{9(81 - 1)}$$

$$= 1 - \frac{432}{720}$$

$$= 1 - 0.6 = 0.4$$

Q-2. 10 competitors in a beauty contest were ranked by three judges. Use the method of rank co-relⁿ to determine which pair of judges has the nearest approach to comm. taste in a

R_1	First Judge	R_2 S.J	R_3 T.J.	R_{12}	D_{12}^2	R_{23}	D_{23}^2	
10	1	8	3	6	-2	4	-3	9
5	6	6	5	4	1	1	1	1
6	5	3	8	9	-3	9	-1	1
1	10	7	4	8	6	36	-4	16
8	3	4	7	1	-4	16	6	36
9	2	1	10	2	-8	64	8	64
7	4	9	2	3	2	4	-1	1
2	9	10	1	10	8	64	-9	81
4	7	5	6	5	1	1	1	1
3	8	2	9	7	-1	1	2	4
					200		214	

$$R_n = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 200}{10(100 - 1)} = 1 - \frac{1200}{990}$$

$$= 1 - 1.212 = -0.212$$

$$R_{23} = 1 - \frac{6 \times 214}{990}$$

$$= 1 - \frac{1284}{990}$$

$$= 1 - 1.297 = -0.297$$

$$R_{31} = 1 - \frac{6 \times 60}{990} = 1 - \frac{360}{990} = 1 - 0.363 = 0.637$$

Spearman's coeff of correlⁿ.
(when there are tied ranks).

$$r = 1 - \frac{6 \left[\sum D^2 + \frac{m_1(m_1^2 - 1)}{12} + \frac{m_2(m_2^2 - 1)}{12} + \dots \right]}{n(n^2 - 1)}$$

where,

$m_1 \rightarrow$ repeated no. of times data either in x or y.

$m_2 \rightarrow$ repeated no. of times second data either in x or y.

Q.1 Find the coeff. of rank co-relⁿ from foll. data.

x	y	R ₁	R ₂	D	D ²	
68	62	4	5	-1	1	$m_1 = 2$ $m_2 = 3$ $m_3 = 2$
64	58	6	7	-1	1	
75	68	2.5	3.5	-1	1	
50	45	9	10	-1	1	
64	81	6	1	5	25	
80	60	1	6	-5	25	
75	68	2.5	3.5	-1	1	
40	48	10	9	1	1	
55	50	8	8	0	0	
64	70	6	2	4	16	
					72	

$$r = 1 - \frac{6 \left[72 + \frac{2(4-1)}{12} + \frac{3(9-1)}{12} + \frac{2(4-1)}{12} \right]}{10(100 - 1)}$$

$$= 1 - \frac{6 \left[72 + 0.5 + 2 + 0.5 \right]}{990} = 1 - \frac{450}{990} = 1 - 0.4545 = 0.545$$

Note - If we apply change of origin & scale in the data set, then Karl Pearson's coeff. of co-relⁿ doesn't change.

$$r_{xy} = r_{uv} = \frac{n\sum uv - \sum u \sum v}{\sqrt{n\sum u^2 - (\sum u)^2} \sqrt{n\sum v^2 - (\sum v)^2}}$$

where $u = \frac{x-a}{h}$, $v = \frac{y-b}{h}$

Q- 10 students got the foll. % marks in the 1st & 2nd state. Calculate the coeff. of co-relⁿ by shifting of or. & ch. of scale (Pearson).

Roll.	Ex(x)	Stat(y)	$u = \frac{x-65}{10}$	$v = \frac{y-65}{10}$	uv	u^2	v^2
1	78	84	13	18	234	169	324
2	36	51	-29	-15	435	841	225
3	98	91	33	25	825	1089	625
4	25	60	-40	-6	240	1600	36
5	75	68	10	2	20	100	4
6	82	62	17	-4	68	289	16
7	90	86	25	20	500	625	400
8	62	58	-3	-8	24	9	64
9	65	53	0	-13	0	0	169
10	39	47	-26	-19	494	676	361
			0	0	2704	5398	2271

$\bar{x} = 65$

$\bar{y} = 66$

$$r_{uv} = \frac{n\sum uv - \sum u \sum v}{\sqrt{n\sum u^2 - (\sum u)^2} \sqrt{n\sum v^2 - (\sum v)^2}}$$

$$= \frac{10(2704) - 0}{\sqrt{10(5398) - 0} \sqrt{10(2271) - 0}} = \frac{27040}{34635} = 0.78$$

$r_{uv} = 0.78$

$$= \frac{27040}{\sqrt{53980} \sqrt{22240}}$$

$$= \frac{27040}{232.3 \times 149.1} = \frac{27040}{34635}$$

$= 0.78$

Regression analysis In this, we obtain the relationship b/w the dependent var. (o/p) and the one or more independent var. (i/p).

The basic differences b/w the curve fitting & the regression analysis is that we can take any var. independent or dependent in the regression var.

$x \rightarrow$ Investment $\rightarrow$ data(x)
 $y \rightarrow$ Profit $\rightarrow$ y(output)

Regression lines - Let we have data set -

$x \rightarrow x_1, x_2, x_3, \dots, x_n$

$y \rightarrow y_1, y_2, y_3, \dots, y_n$

Case-I (Regression line y on x) $x \rightarrow$ Ind, $y \rightarrow$ dep

Here we take x as independent var. -

$y = a + bx$ — (1)

Apply method of least sq. (Alom. eqn)

$$y = a + bx$$

$$\sum y = a + b \sum x \quad \text{--- (i)}$$

$$\sum xy = a \sum x + b \sum x^2 \quad \text{--- (ii)}$$

On solving (i) & (ii): we get a & b.
Hence we get reg. line.

$$y = a + bx$$

OR

Regression line y on x:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where

b_{yx} = regression coeff.
y on x.

where b_{yx} is given by-

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

Ques-II (Regression line x on y).

Let regression line x on y-

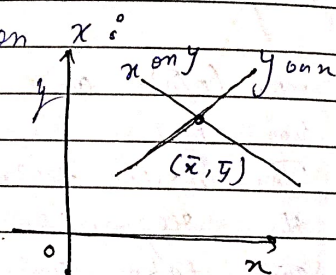
$$x = c + dy \quad \text{--- (i)}$$

Apply method of least sq.
normal eqn.

$$\sum x = c \sum 1 + d \sum y \quad \text{--- (ii)}$$

$$\sum xy = c \sum y + d \sum y^2 \quad \text{--- (iii)}$$

On solving, we get c & d.
If put in (i), we get -



required regression line x on y.

OR

Regression line x on y:

$$x - \bar{x} = \frac{b_{xy}}{\text{slope}} (y - \bar{y})$$

Note -

$$b_{yx} = r \frac{\sigma_x}{\sigma_y}$$

$$b_{xy} = r \frac{\sigma_y}{\sigma_x}$$

$b_{yx} \rightarrow$ reg. coeff. x on y
 $r \rightarrow$ (pearson) coef of co-ord.
 $\sigma_y \rightarrow$ S.D of y
 $\sigma_x \rightarrow$ S.D of x

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum y^2 - (\sum y)^2}}$$

(±)

$$r = \sqrt{b_{xy} \times b_{yx}}$$

(Q.P)

Here b_{xy} , b_{yx} , & r have same sign.

1. Obtain the both regression lines for the foll. data set

x	y	xy	x ²	y ²
1.	3	3	1	9
2.	7	14	4	49
3.	10	30	9	100
4.	12	48	16	144
5.	14	70	25	196
6.	17	102	36	289
7.	20	140	49	400
8.	24	192	64	576
36	107	599	204	1763

136.5



$$y = a + bx \quad \text{--- (I)}$$

Normal eqn are -

$$\Sigma y = a \Sigma 1 + b \Sigma x \quad \text{--- (II)}$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \quad \text{--- (III)}$$

$$107 = 8a + 36b \quad \times 9$$

$$963 = 36a + 204b \quad \times 2$$

$$a = 0.785$$

$$b = 2.797$$

$$963 = 72a + 36b$$

$$1198 = 72a + 408b$$

$$235 = 36b \Rightarrow b = 2.797$$

$$a = 0.785$$

Regr. line y on x is

$$y = 0.785 + 2.797x$$

Now, let reg. line x on y be

$$x = c + dy$$

Normal eqn are -

$$\Sigma x = c \Sigma 1 + d \Sigma y \quad \text{--- (IV)}$$

$$\Sigma xy = c \Sigma y + d \Sigma y^2 \quad \text{--- (V)}$$

$$36 = 8c + 107d$$

$$963 = 107c + 1763d$$

$$c = -0.235$$

$$d = 0.354$$

$$c = -0.235$$

$$d = 0.354$$

x =



Properties of coeff. of regression.

- i) The co-relⁿ coeff is the geometric mean of the two regression coeff b_{xy} & b_{yx} .

$$r = \pm \sqrt{b_{xy} \times b_{yx}}$$

- ii) If one of the reg. coeff is greater than unity, the other reg. coeff. must be less than unity.

- iii) The arithmetic mean of the two reg. coeff. is greater than the co-relⁿ coeff.

$$\frac{b_{yx} + b_{xy}}{2} > r$$

- iv) The reg. coeff. are independent of the change of origin or change of scale.

- v) The co-relⁿ coeff and the two reg. coeff. have same sign.

- Q - If the two coeff. of reg. are 0.4 & 1.6 then find the coeff. of co-relⁿ.

$$r = \pm \sqrt{b_{xy} \times b_{yx}}$$

$$= \sqrt{0.4 \times 1.6}$$

$$r = \sqrt{0.64}$$

$$= 0.8$$

$\because b_{xy}, b_{yx}$ & r must be same s.

Q.1- When the regression lines are in the form as:

$$\begin{aligned} y \text{ on } x & \rightarrow a_1x + b_1y = c_1 \quad (a_1x + b_1y - c_1 = 0) \\ x \text{ on } y & \rightarrow a_2x + b_2y = c_2 \quad (a_2x + b_2y - c_2 = 0) \end{aligned}$$

I- When the two regr. lines are in the form as.

$$\begin{aligned} C_1y &= a_1x + b_1 \rightarrow y \text{ on } x \\ C_2x &= a_2y + b_2 \rightarrow x \text{ on } y \end{aligned}$$

Q.3 In a attack of WTC destroyed lab record of an analysis of the co-relⁿ of regr. data, the following results are left only,

Variance of $x = 9$

Regression lines are - $8x - 10y + 66 = 0$ (y on x)
 $40x - 18y = 214$ (x on y)

Determine i) The mean value of data x and y

ii) The coeff. of co-relⁿ are.

iii) Std. deviation of y

$$x = 13$$

$$y = 17$$

$$8x - 10y + 66 = 0$$

$$40x - 18y = 214$$

$$x = 13, y = 17$$

$$\bar{x} = 13, \bar{y} = 17$$

ii) from (i):

$$8x - 10y + 66 = 0$$

$$10y = 8x + 66$$

$$y = \frac{8}{10}x + \frac{66}{10}$$

$$\text{Hence by } \frac{8}{10} = 0.8$$

from (ii):

$$40x - 18y = 214$$

$$40x = 18y + 214$$

$$x = \frac{18}{40}y + \frac{214}{40}$$

Hence,

$$b_{xy} = \frac{18}{40} = 0.45$$

But

$$r = \pm \sqrt{b_{yx} \times b_{xy}}$$

$$= \pm \sqrt{0.8 \times 0.45} = 0.6$$

$$r = \pm 0.6$$

iii)

Given-

$$\sigma_x^2 = 9$$

$$\sigma_x = \pm 3$$

We know that

$$b_{yx} = \frac{\sigma_y}{\sigma_x}$$

$$0.8 = 0.6 \times \frac{\sigma_y}{3}$$

$$\sigma_y = \frac{8 \times 3}{6} = 4$$

$$\sigma_y = 4$$

Non linear regression curve -

We calculate both regression curve x on y and y on x by the method of least squares of a given data